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Appendix-A1 Example Track Data TDM Files

A1.1 Example Track Mode 1 TDM File

This example shows a TDM file collected using mode 1: ranging only. Ranging values are timestamped to 1 us resolution, and range delay values are reported in units of seconds, with a resolution of 1 ps.

File Name: TRACK_D32_1964-123A_2005-10-16T11-12-23.tdm

```
CCSDS_TDM_VERS = 2.0
COMMENT Ground antenna: H16 Aurora, Colorado, USA
COMMENT WGS84 coordinates: lat=39.613621, long=-104.675963, alt=1813.56
COMMENT ECEF coordinates: X=-1246890.058, Y=-4760998.779, Z=4046185.863
COMMENT Pedestal offset: Lg=4.331230 meters
COMMENT X-band TLT calibration date: 15 NOV 2023
COMMENT Spacecraft: Apollo 42
COMMENT COSPAR: 1964-123A Catalog: 987654321
CREATION_DATE = 2005-10-16T11:12:23
ORIGINATOR = KSAT

META_START
COMMENT Range Tracking Data: Mode 1=range only
TIME_SYSTEM = UTC
PARTICIPANT_1 = H16
PARTICIPANT_2 = 1964-123A
MODE = SEQUENTIAL
TRANSMIT_BAND = X
RECEIVE_BAND = X
INTEGRATION_INTERVAL = 1.0
INTEGRATION_REF = END
RANGE_UNITS = s
TRANSMIT_DELAY_1 = 0.000001023
```

A1.3 Example Track Mode 3 TDM File

Appendix-A1 Example Tra...

A1.1 Example Track Mode 1...

A1.3 Example Track Mode ...

A1.4 Example Track Mode ...

This example shows a TDM file collected using mode 3: range+range rate. Range values are reported as in example 1. Uplink and downlink frequency values are reported as in example 4.

File Name: TRACK_D32_1964-123A_2005-10-16T11-12-23.tdm

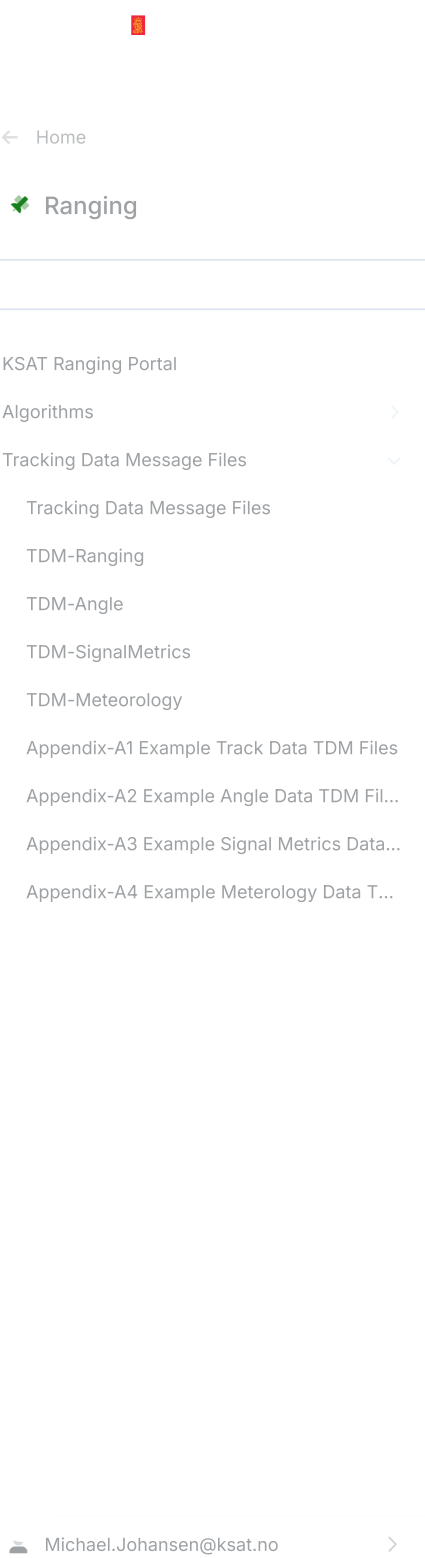
TRANSMIT_FREQ_1	=	2005-10-16T11:12:23.987654	/1900000000.000000
RECEIVE_FREQ_1	=	2005-10-16T11:12:23.987654	8447530040.000000
RANGE	=	2005-10-16T11:12:24.987641	998.515676770200
RECEIVE_FREQ_1	=	2005-10-16T11:12:24.987641	8447529960.357520
RANGE	=	2005-10-16T11:12:25.987628	998.516154898100
RECEIVE_FREQ_1	=	2005-10-16T11:12:25.987628	8447529880.715040
RANGE	=	2005-10-16T11:12:26.987615	998.516633026000
RECEIVE_FREQ_1	=	2005-10-16T11:12:26.987615	8447529801.072570
RANGE	=	2005-10-16T11:12:27.987602	998.517111153900
RECEIVE_FREQ_1	=	2005-10-16T11:12:27.987602	8447529721.430090
RANGE	=	2005-10-16T11:12:28.987589	998.517589281800
RECEIVE_FREQ_1	=	2005-10-16T11:12:28.987589	8447529641.787610
RANGE	=	2005-10-16T11:12:29.987576	998.518067409700
RECEIVE_FREQ_1	=	2005-10-16T11:12:29.987576	8447529562.145130
RANGE	=	2005-10-16T11:12:30.987563	998.518545537600
RECEIVE_FREQ_1	=	2005-10-16T11:12:30.987563	8447529482.502650
RANGE	=	2005-10-16T11:12:31.987550	998.519023665500
RECEIVE_FREQ_1	=	2005-10-16T11:12:31.987550	8447529402.860180
RANGE	=	2005-10-16T11:12:32.987537	998.519501793400
RECEIVE_FREQ_1	=	2005-10-16T11:12:32.987537	8447529323.217700
RANGE	=	2005-10-16T11:12:33.987524	998.519979921300
RECEIVE_FREQ_1	=	2005-10-16T11:12:33.987524	8447529243.575220
RANGE	=	2005-10-16T11:12:34.987511	998.520458049200
RECEIVE_FREQ_1	=	2005-10-16T11:12:34.987511	8447529163.932740

A1.4 Example Track Mode 4 TDM File

This example shows a TDM file collected using mode 4: range rate only. Transmit and receive frequencies are each reported to a 1uHz resolution. The transmit frequency is the frequency defined in the mission file (assuming that Doppler pre-compensation is not used on uplink). The receive frequency is the nominal receive frequency defined in the mission file, corrected for the Doppler shift measured in the carrier tracking loop of the receiver.

File Name: TRACK_D32_1964-123A_2005-10-16T11-12-23.tdm

```
CCSDS_TDM_VERS = 2.0
COMMENT Ground antenna: H16 Aurora, Colorado, USA
COMMENT WGS84 coordinates: lat=39.613621, long=-104.675963, alt=1813.56
COMMENT ECEF coordinates: X=-1246890.058, Y=-4760998.779, Z=4046185.863
COMMENT Pedestal offset: Lg=4.331230 meters
COMMENT X-band TLT calibration date: 15 NOV 2023
```



COMMENT Spacecraft: Apollo 42
COMMENT COSPAR: 1964-123A Catalog: 987654321
CREATION_DATE = 2005-10-16T11:12:23
ORIGINATOR = KSAT

```
META_START
COMMENT Range Tracking Data: Mode 4=Doppler only
TIME_SYSTEM = UTC
PARTICIPANT_1 = H16
PARTICIPANT_2 = 1964-123A
MODE = SEQUENTIAL
TRANSMIT_BAND = X
RECEIVE_BAND = X
TURNAROUND_NUMERATOR = 880
TURNAROUND_DENOMINATOR = 749
INTEGRATION_INTERVAL = 1.0
INTEGRATION_REF_FSR
```

